

CULTURALLY AWARE AI

MARIEKE VAN ERP

AI IS USEFUL

safer driving experience



hypothesis generation

BY JENNIFER DOUDNA SCIENCE NOV 26, 2024 5:10 AM

Combining AI and Crispr Will Be Transformational

The genome-editing technology can be supercharged by artificial intelligence—and the results are already being felt.

personalised medicine

AI-powered personalised medicine could revolutionise healthcare (and no, we're not putting ChatGPT in charge)

Mihaela van der Schaar

Artificial intelligence can't replace human professionals but it could transform the way they treat diseases such as cancer, and save lives

● Mihaela van der Schaar is director of the Cambridge Centre for AI in Medicine at the University of Cambridge



nalise-medicine-patient-lab-health-diagnosis-cambridge#img-1

AI HAS PROBLEMS

AI has a copyright problem

n p r

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BUSINESS

'The New York Times' takes OpenAI to court. ChatGPT's future could be on the line

UPDATED JANUARY 14, 2025 · 4:27 PM ET ⓘ



Bobby Allyn



A sign for The New York Times hangs above the entrance to its building, Thursday, May 6, 2021, in New York. The New York Times filed a federal lawsuit against OpenAI and Microsoft on Wednesday, Dec. 27, 2023, seeking to end the practice of using published material to train chatbots.

Mark Lennihan/AP Photo

AI systems need a lot of energy

IMF study shows concern — but not panic — over AI's climate toll



Ben Geman



Illustration: Natalie Peebles/Axios

A new International Monetary Fund study of the AI-climate-energy nexus finds reason for worry, but hardly panic.

Why it matters: Gaming out artificial intelligence's energy needs and the emissions in tow is a big challenge for policymakers, tech companies and power providers.

What they found: Under current energy policies, the IMF projects a cumulative 1.7 gigatons of additional CO₂ emissions linked to AI's needs from 2025-2030.

AI systems can exhibit ethically dubious behaviour

https://www.vice.com/en/article/v7dg8m/ethical-ai-trained-on-reddit-posts-said-genocide-is-okay-if-it-makes-people-happy?utm_source=

VICE

Ethical AI Trained on Reddit Posts Said Genocide Is OK If It Makes People Happy

Fed by Reddit and Mechanical Turks, the algorithmic morality spinner is a dangerous and stupid use of technology.

By [Matthew Gault](#)

November 3, 2021, 4:58pm [Share](#) [Tweet](#) [Snap](#)



IMAGE: GETTY IMAGES

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AI systems can show biases



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Why Amazon's Automated Hiring Tool Discriminated Against Women



Rachel Goodman,
Former Staff
Attorney,
ACLU Racial Justice
Program

Share This Page



October 12, 2018

In 2014, a team of engineers at Amazon began working on a project to automate hiring at their company. Their task was to build an algorithm that could review resumes and determine which applicants Amazon should bring on board. But, according to a [Reuters](#) report this week, the project was canned just a year later, when it became clear that the tool systematically discriminated against women applying for technical jobs, such as software engineer positions.

It shouldn't surprise us at all that the tool developed this kind of bias. The existing pool of Amazon software engineers is overwhelmingly male, and the new software was fed data about those engineers' resumes. If you simply ask software to discover other resumes that

AI systems can exhibit racist behaviour

Computer Vision

Image recognition

Responsible AI

AI Ethics

Sony Scientists Find Computer Vision Models Biased on Race, Skin Color

Study: AI models show racial bias; Sony researchers devise novel multidimensional skin color scale to promote fairness

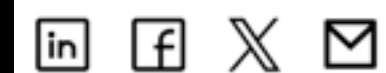


Ben Wodecki, Jr. Editor
October 9, 2023

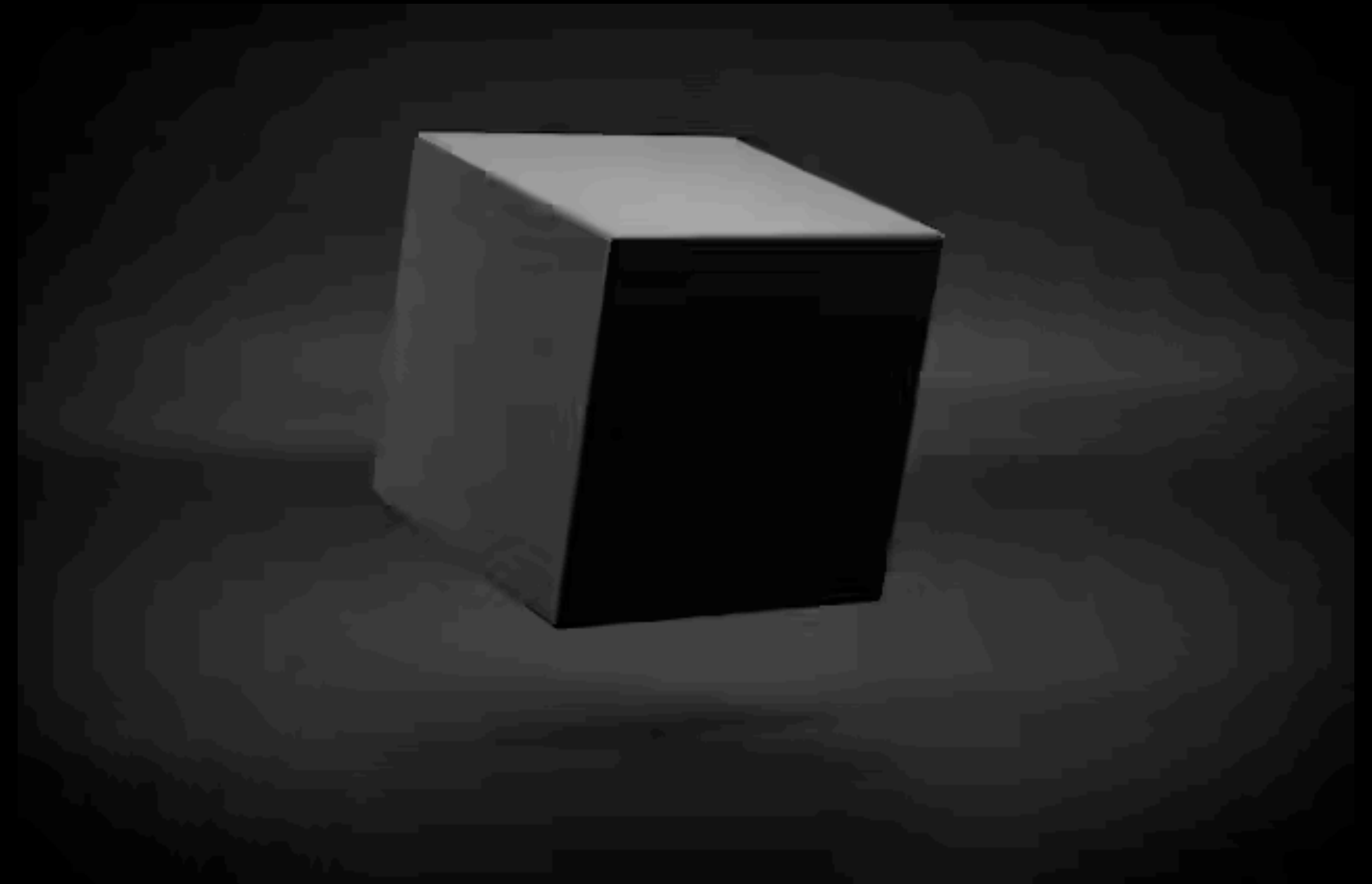
🕒 4 Min Read



SONY



AI is often a black box

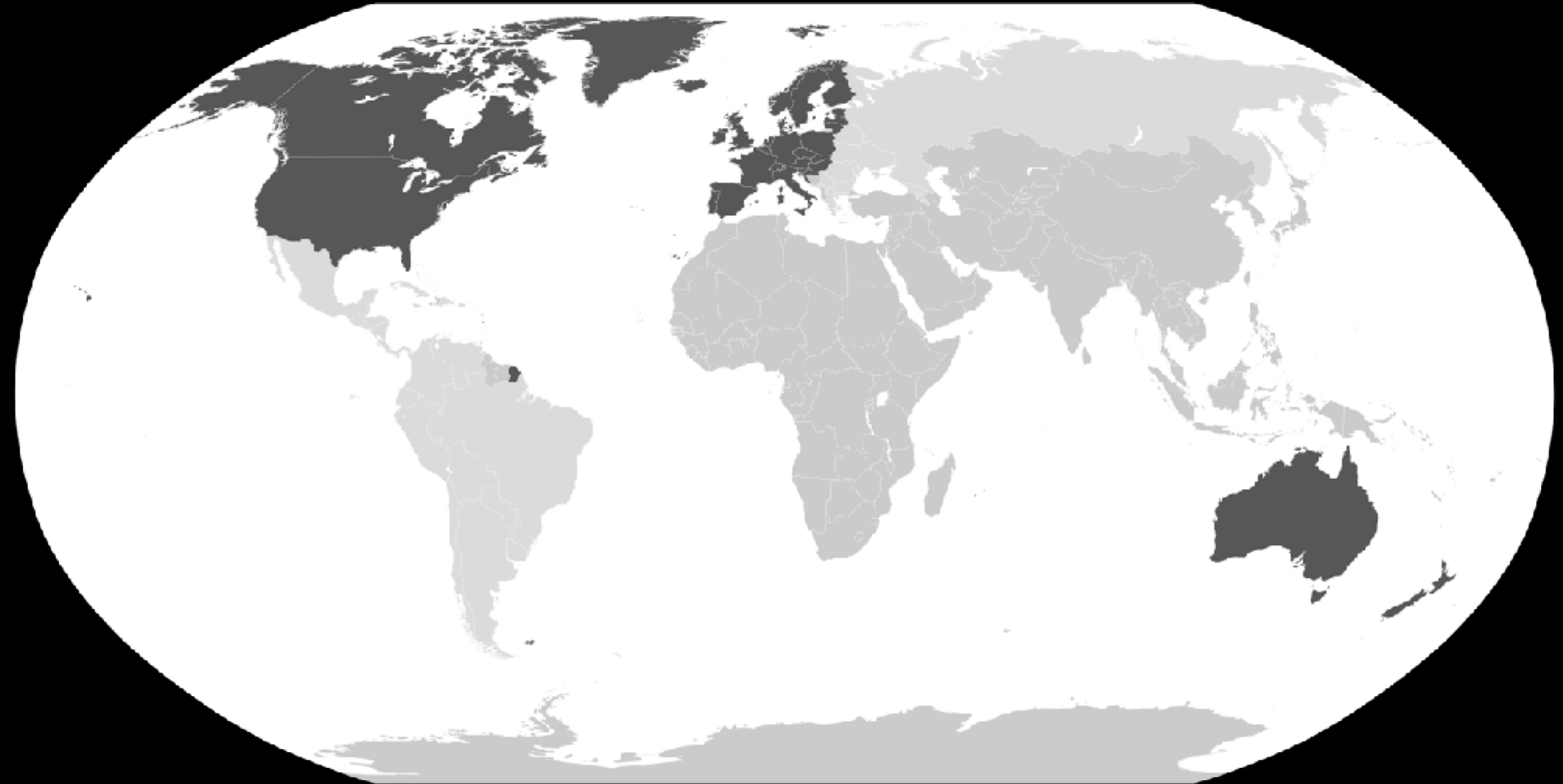


HOW DID THIS HAPPEN?

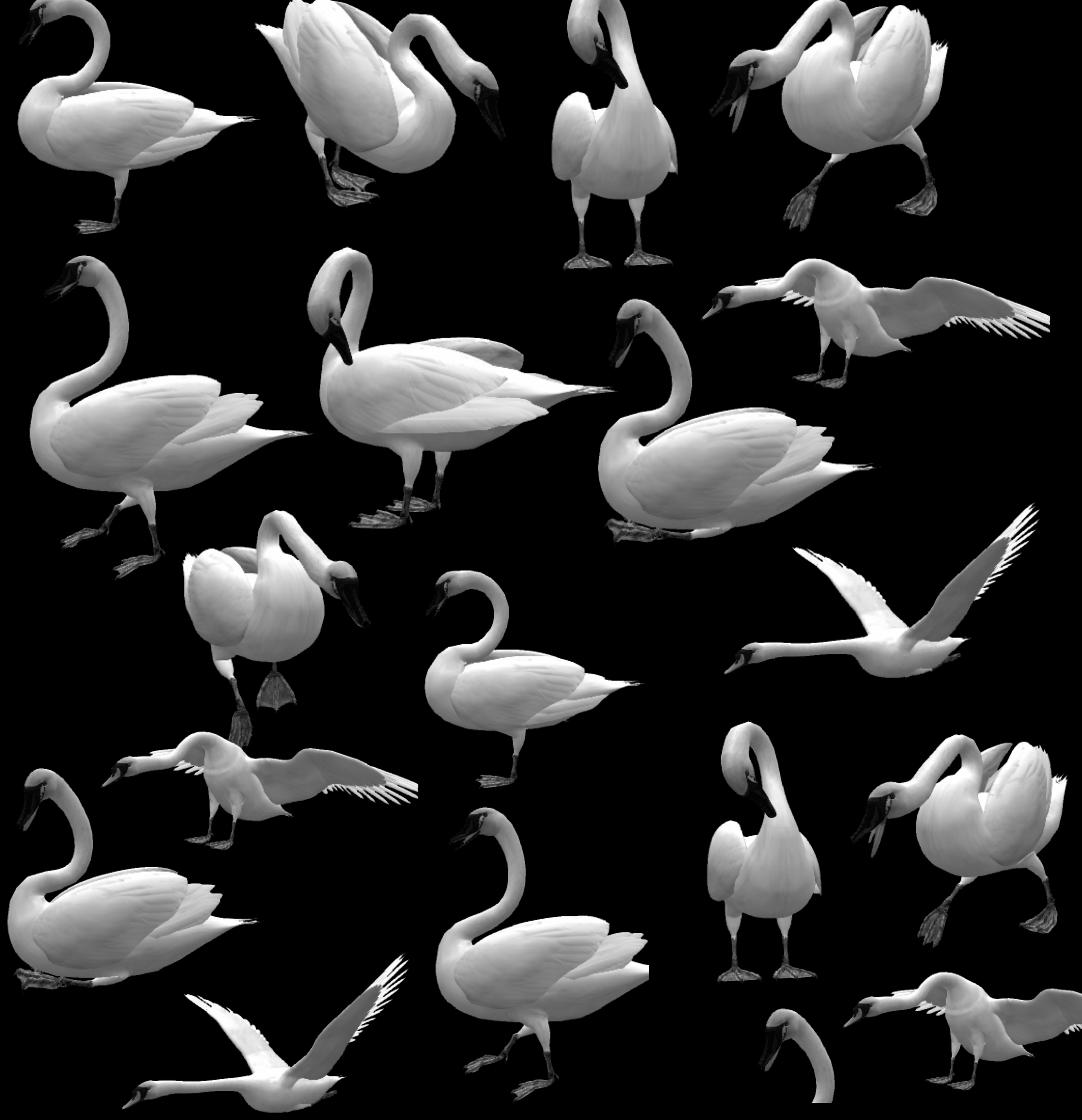
we often only see the tip of the
iceberg



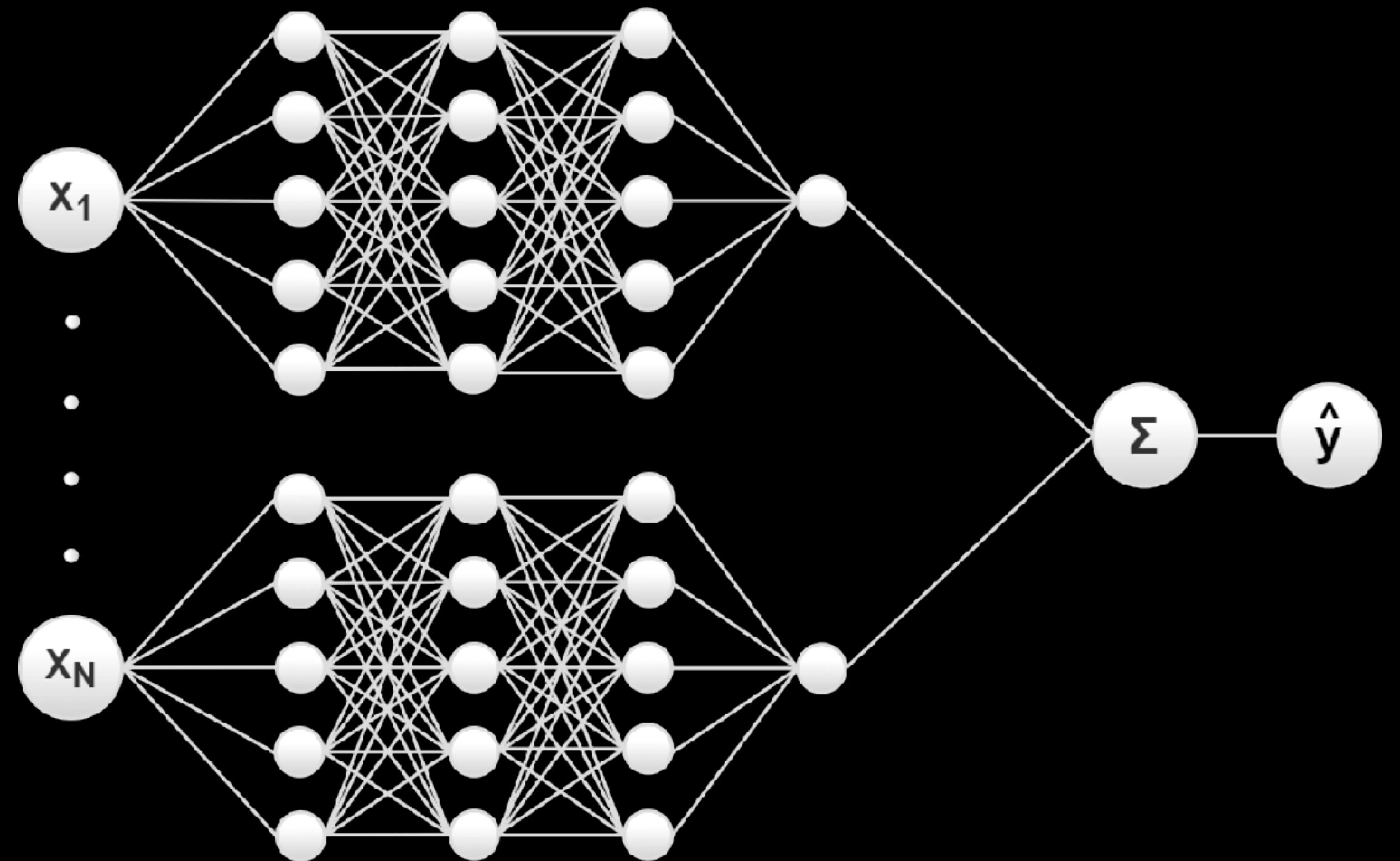
Western AI development induces
Western biases



statistics favours frequency



huge datasets & multilayered
algorithms

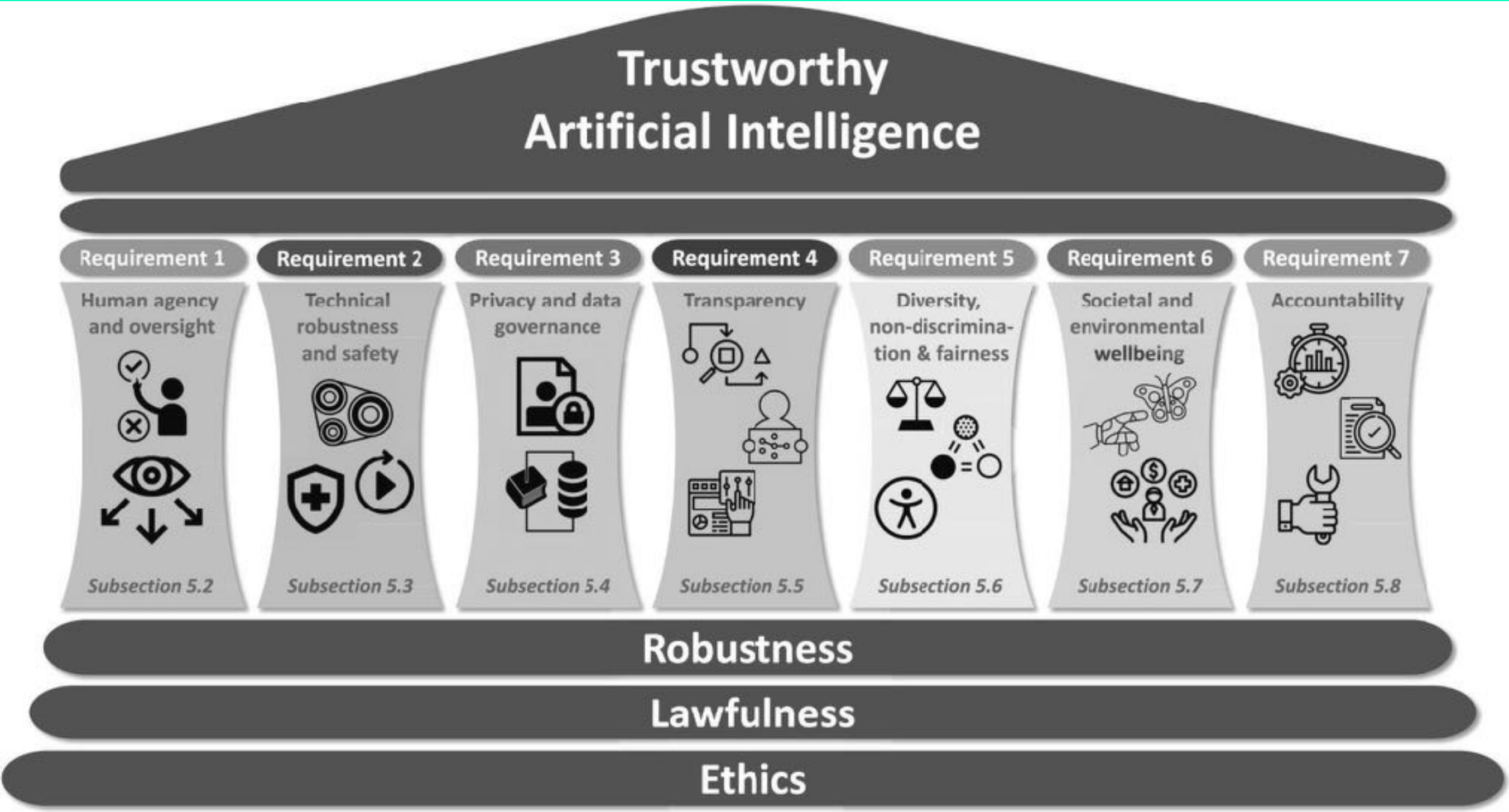


“On two occasions I have been asked, "Pray, Mr. Babbage, if you put into the machine wrong figures, will the right answers come out?" ... I am not able rightly to apprehend the kind of confusion of ideas that could provoke such a question.”

Charles Babbage, Passages from the Life of a Philosopher (1864)

AI CAN BE FIXED

guiding principles for better AI



lawfulness

“KB National Library restricts access to collections for training commercial AI”

NOS Nieuws ▾ Sport ▾ Live Programma's    



NOS Nieuws • Vrijdag 12 januari, 21:04 • Aangepast vrijdag 12 januari, 22:19 

Nationale bibliotheek beperkt toegang tot collecties voor commerciële AI

De KB nationale bibliotheek in Den Haag neemt maatregelen tegen commerciële bedrijven die zonder toestemming haar collectie gebruiken om kunstmatige intelligentie te trainen.

De bibliotheek is bezig haar collectie aan boeken, kranten en andere teksten te digitaliseren. Commerciële bedrijven, zoals het bedrijf achter ChatGPT, gebruiken die bronnen voor het trainen van kunstmatige intelligentie (AI). "De teksten van de KB nationale bibliotheek zijn op hoge kwaliteit gedigitaliseerd, dus ze zijn heel interessant voor die bedrijven", zegt directielid Martijn Kleppe van de bibliotheek in het radioprogramma *Nieuws en Co*.

community-driven AI

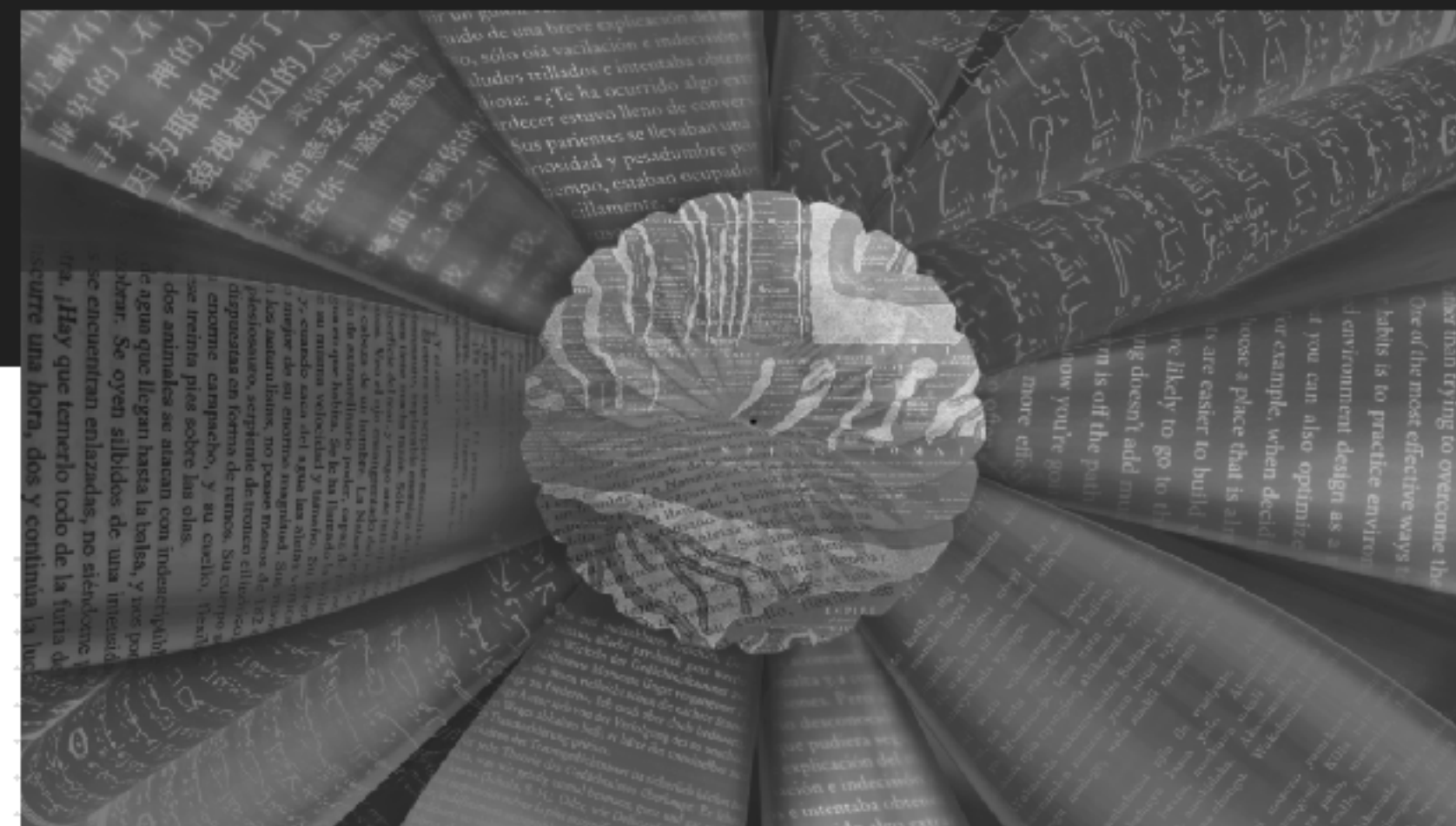
ARTIFICIAL INTELLIGENCE

Inside a radical new project to democratize AI

A group of over 1,000 AI researchers has created a multilingual large language model bigger than GPT-3—and they're giving it out for free.

By Melissa Heikkilä

July 12, 2022



MS TECH | ERVATO, UNSPLASH

PARIS — This is as close as you can get to a rock concert in AI research. Inside the supercomputing center of the French National Center for Scientific Research, on the outskirts of Paris, rows and rows of what look like black fridges hum at a deafening 100 decibels.

They form part of a supercomputer that has spent 117 days gestating a new large language model (LLM) called BLOOM that its creators hope represents a radical departure from the way AI is usually developed.

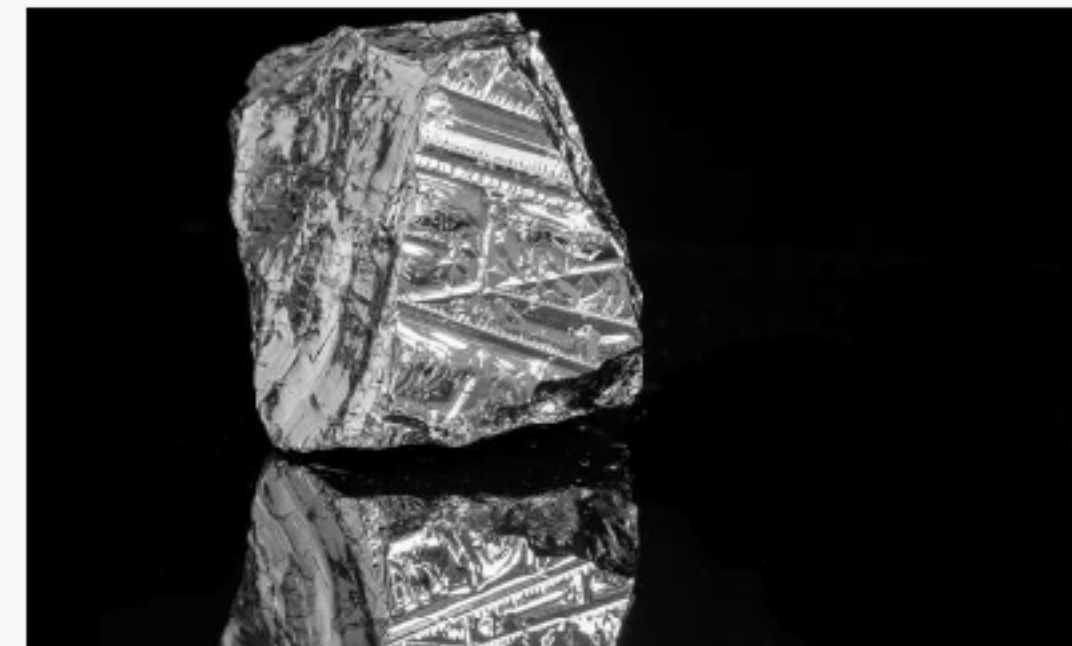
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better datasets

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Images of AI library

Welcome to the inspiration library. These images are just the start of a growing repository, so keep checking in! Images can be downloaded and used by anyone for free if credited using the Creative Commons license referenced on the image card.



Silicon on Black 1 - Catherine Breslin



Quantified Human - Alan Warburton



A trainer instructing a data annotator on how to label images - Nacho Kamenov & Humans in the Loop



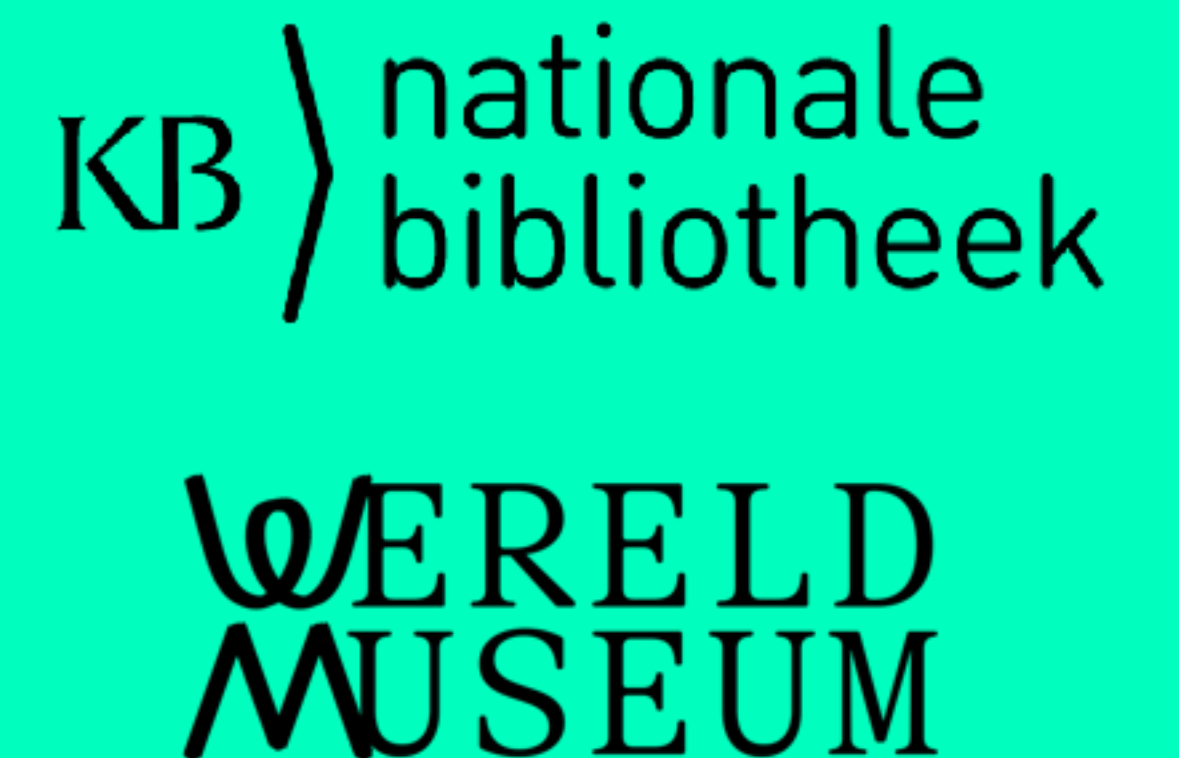
Data flock (faces) - Philipp Schmitt & AT&T Laboratories Cambridge

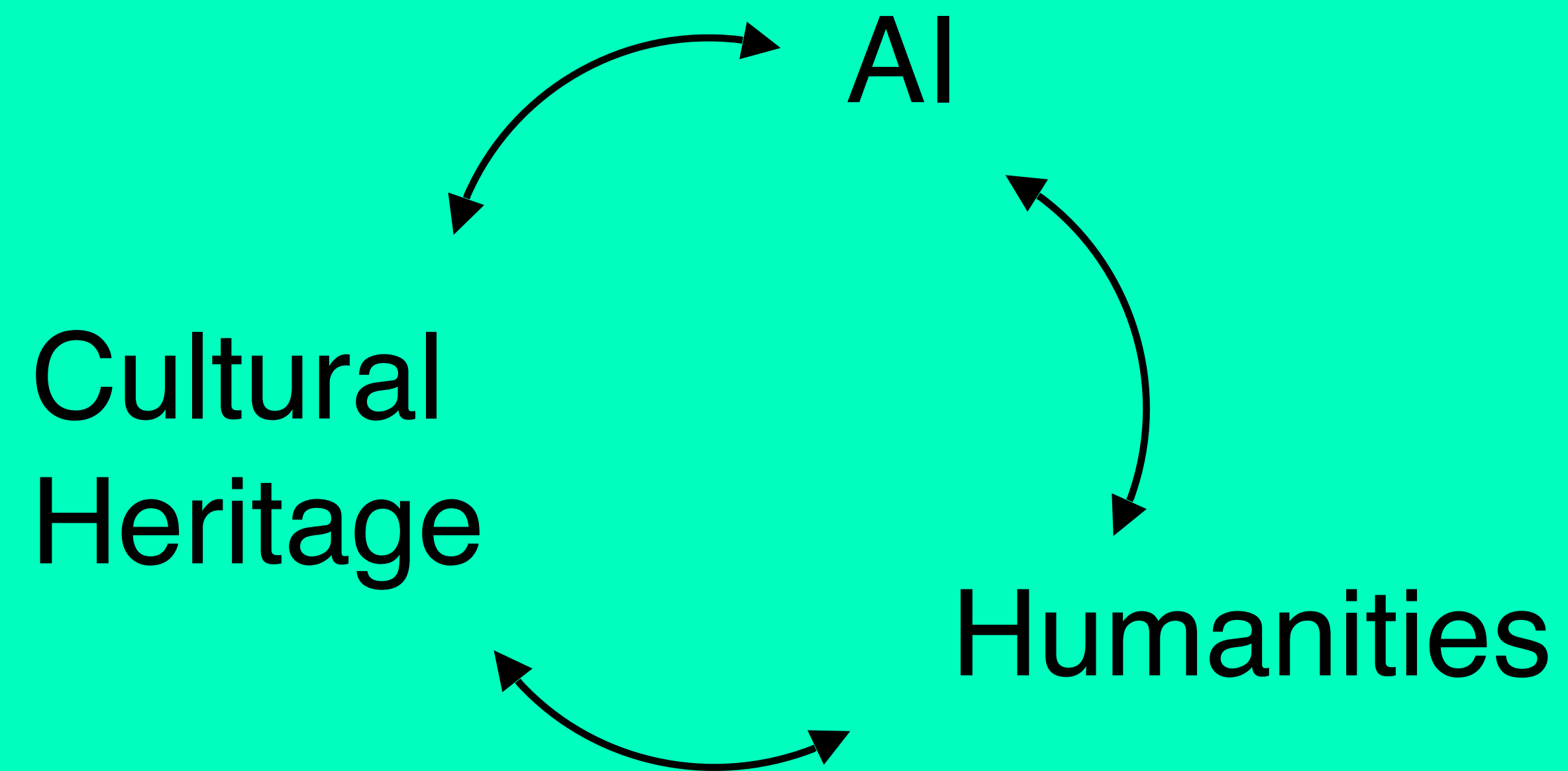


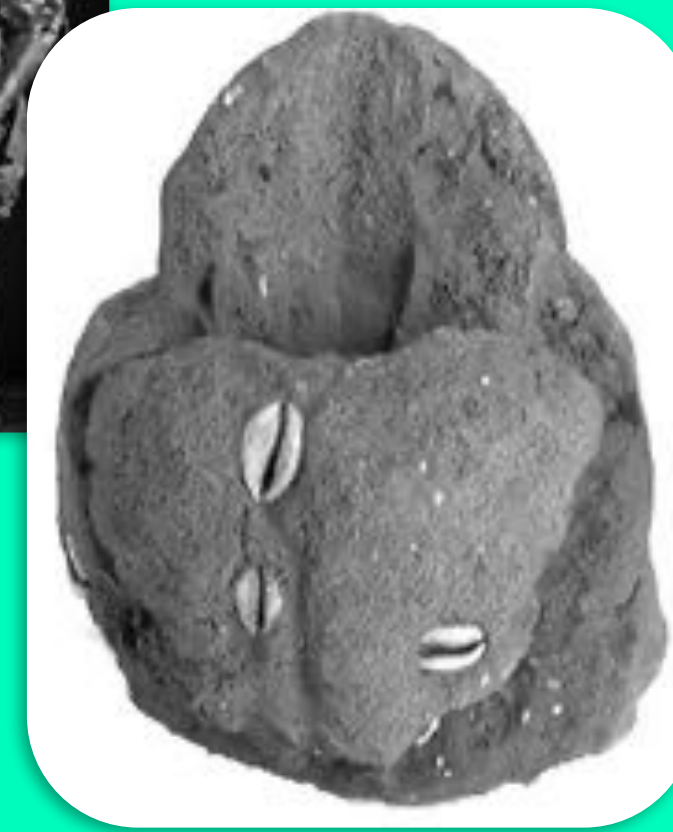
4 research institutes:



4 cultural heritage institutions:





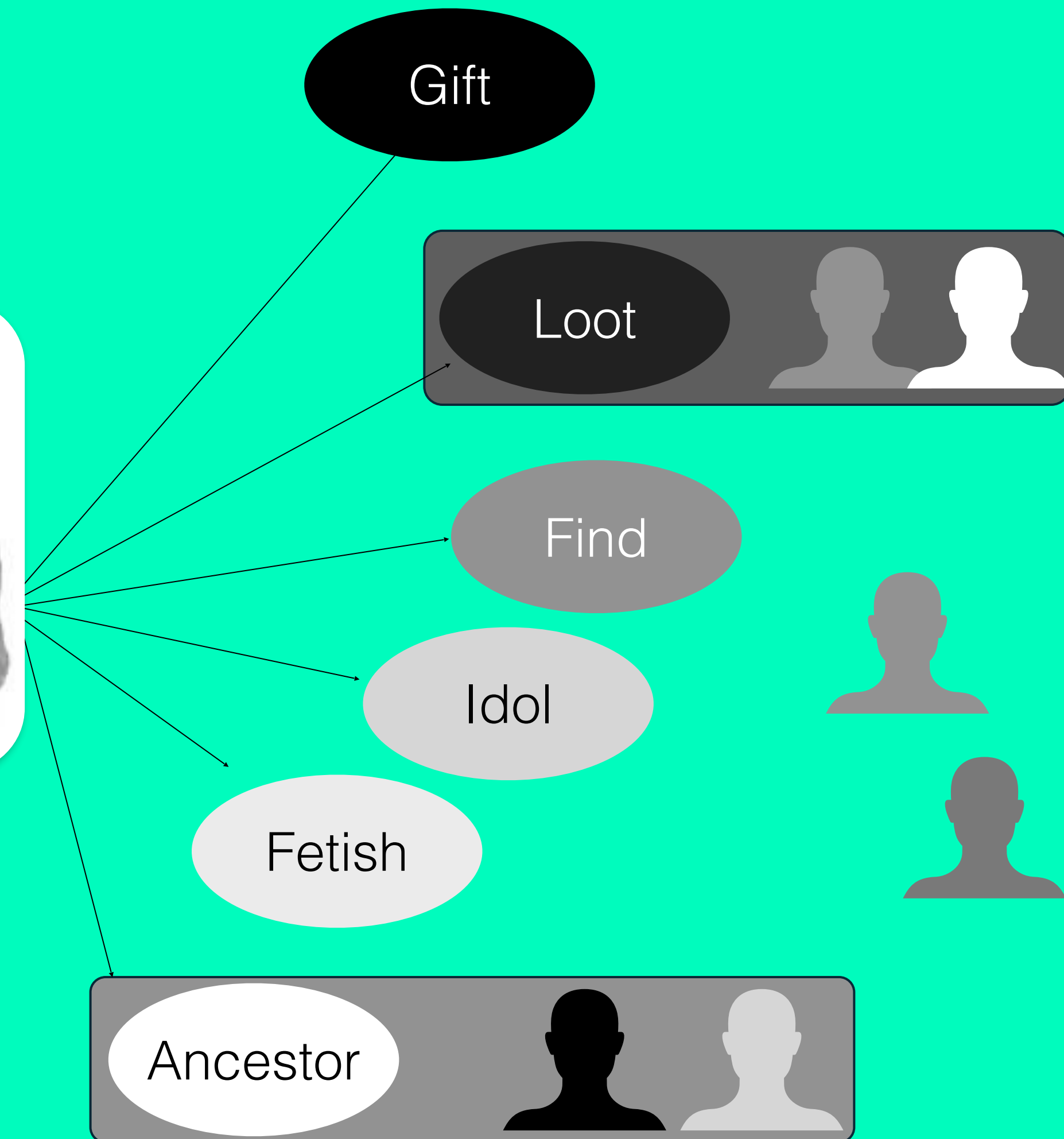


multiple perspectives

what is this object?

what is the purpose of this object?

how should the object be treated?



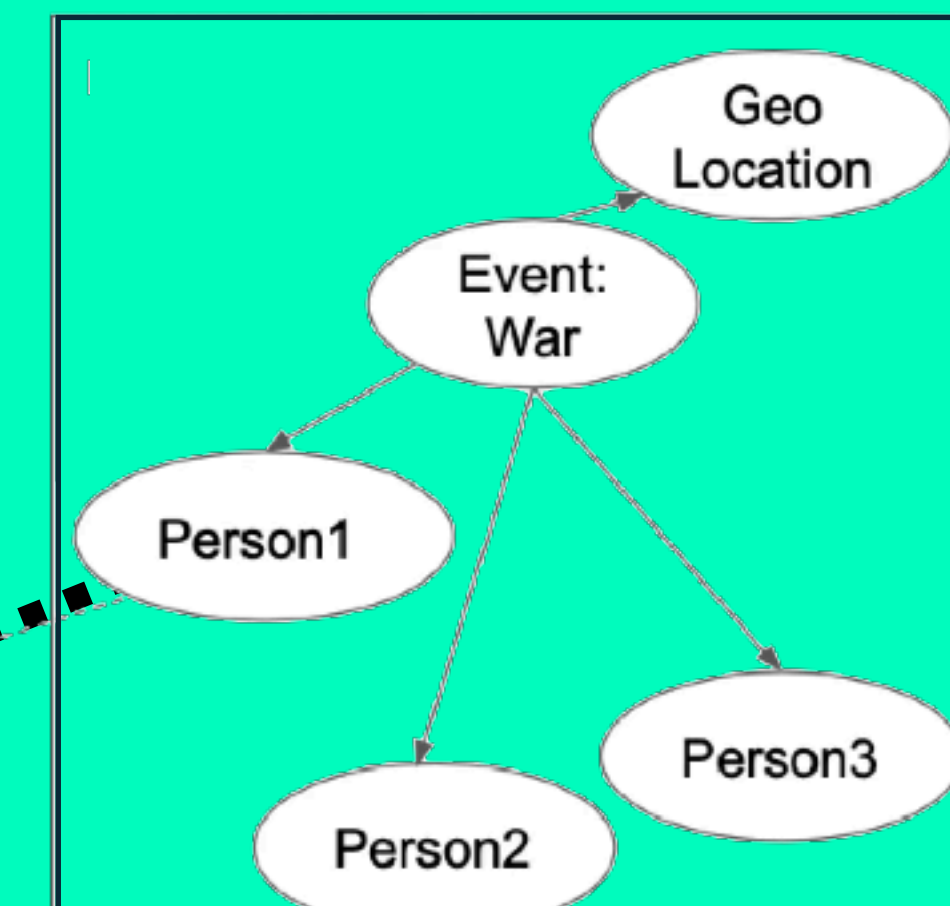
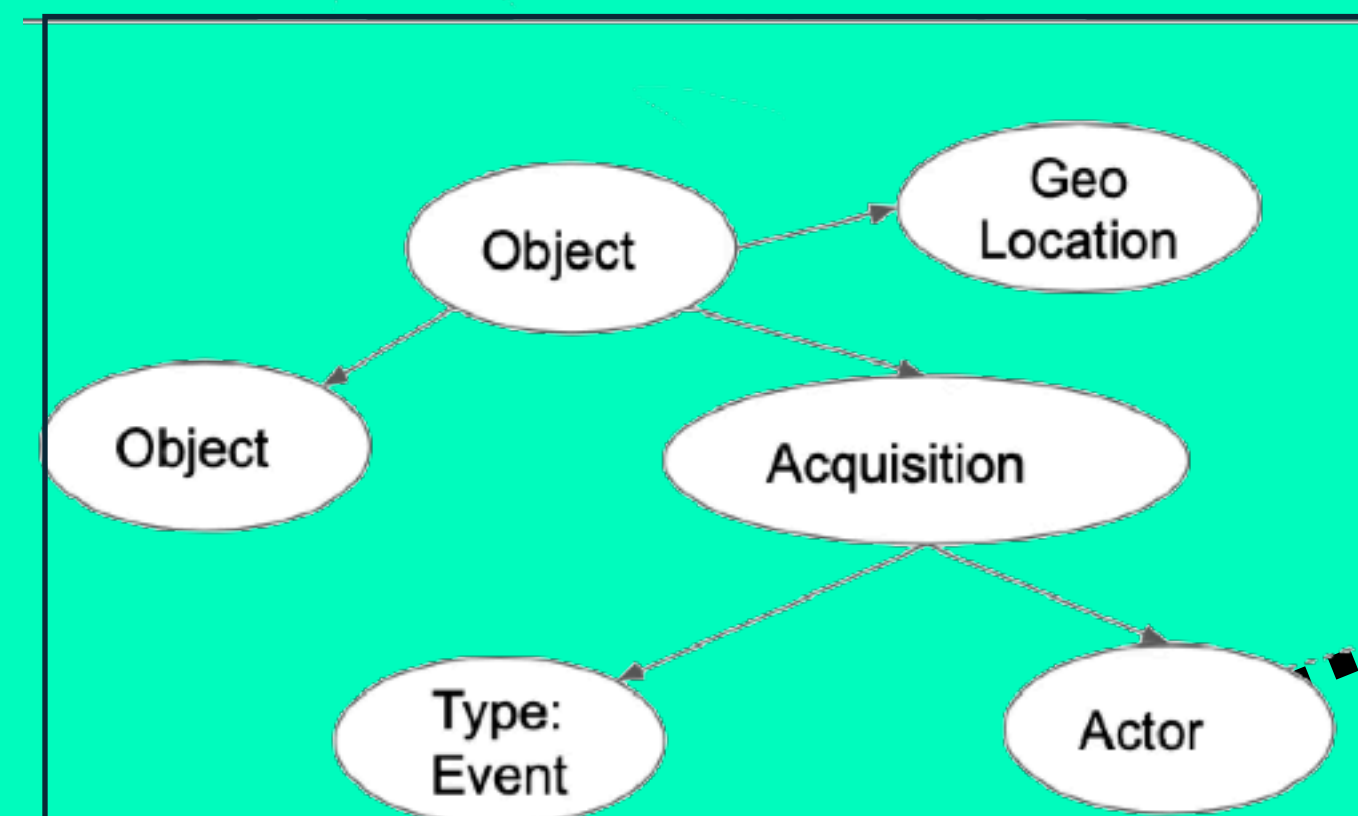


Wereldmuseum



Bronbeek museum

connected narratives across
collections

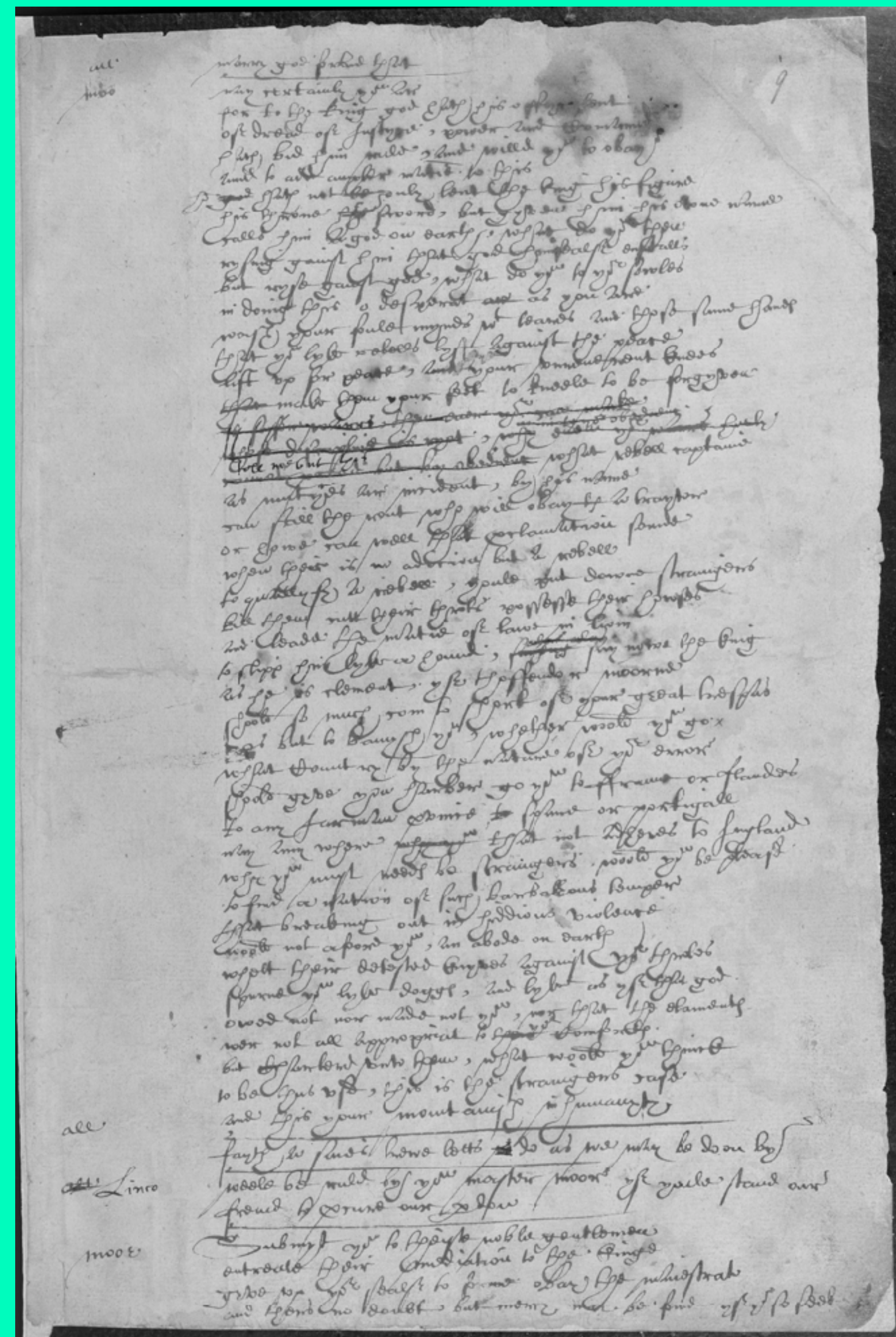


words matter



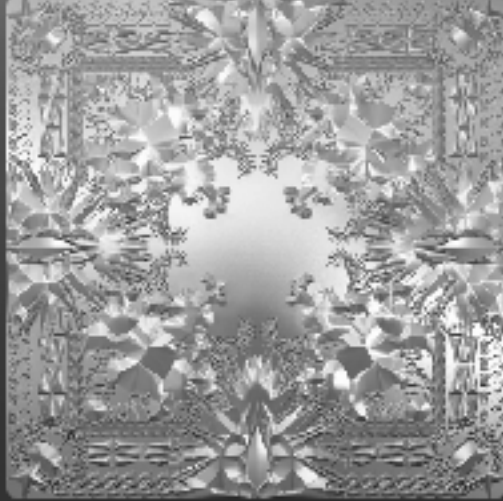
language changes over time

spelling changes
meaning changes



what is acceptable language?

who says what?
in what context?



Song

Ni**as In Paris

 **JAY-Z** • Watch The Throne (Deluxe) • 2011 • 3:39 • 1,613,788,546



Lyrics

We're gonna skate to one song, one song only
Ball so hard, motherfuckers wanna fine me
♪
So I ball so hard, motherfuckers wanna fine me
But first niggas gotta find me
What's fifty grand to a motherfucker like me?
Can you please remind me?
Ball so hard, this shit crazy
Y'all don't know that don't shit faze me
The Nets could go 0 for 82
And I look at you like this shit gravy
Ball so hard, this shit weird
...[Show more](#)



Artist
JAY-Z



Artist
Kanye West

contentious terms in context

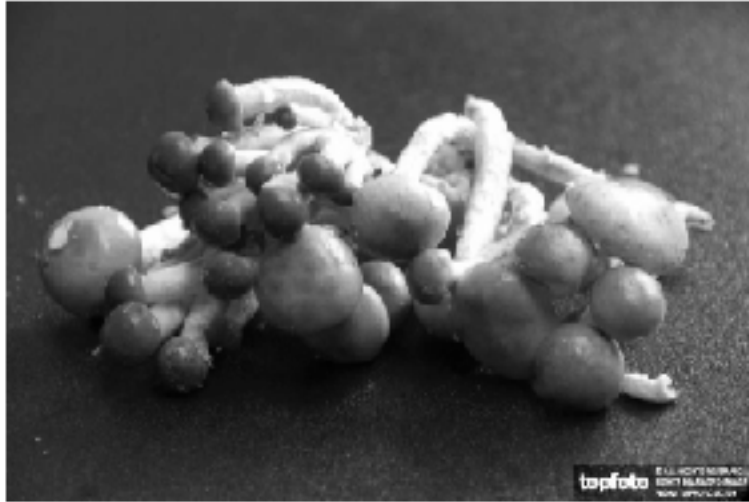
*crowdsourced corpus of
potentially charged terms in
newspaper data*

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Exotic cultivated mushrooms - Cinnamon

Exotic cultivated mushrooms - Cinnamon Cap _x000D_ credit: Marie-Louise Avery / thePictureKitchen / TopFoto

Created by
thePictureKitchen / EUFD

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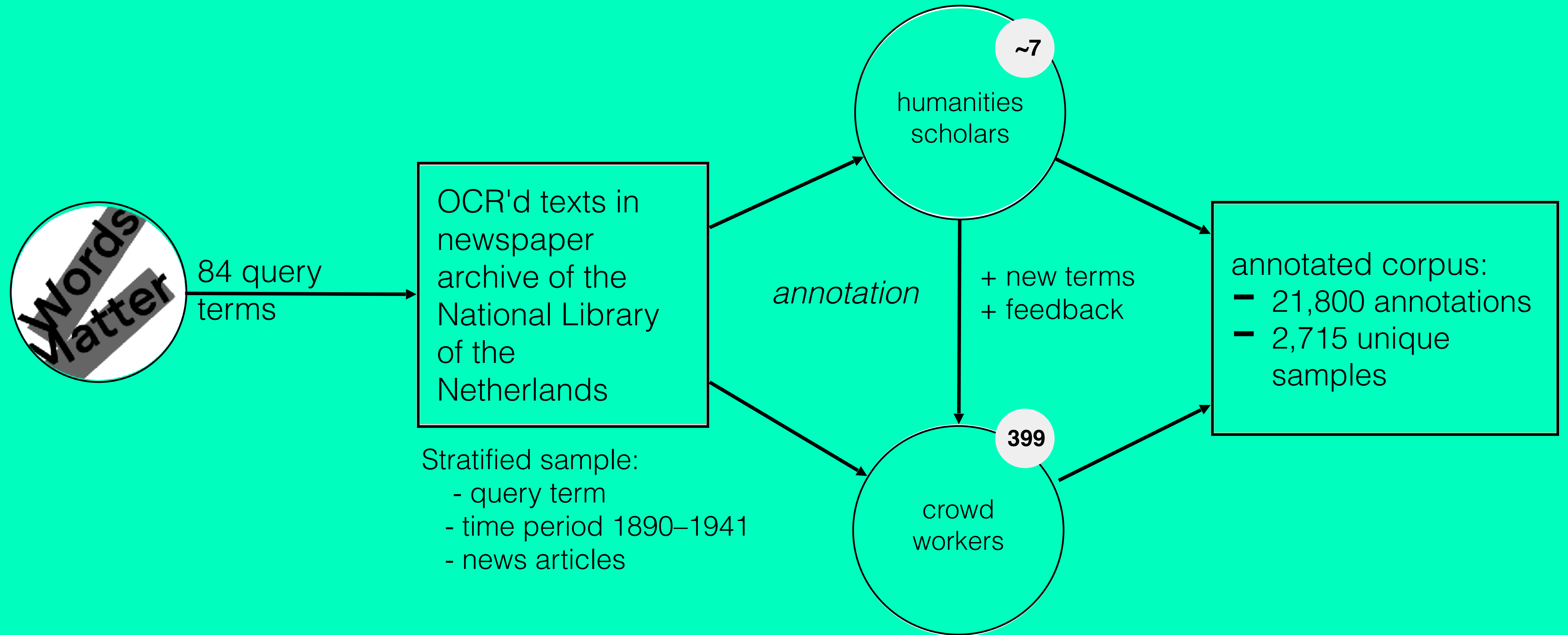


Exotic visitors for London_x000D_ H H

Exotic visitors for London_x000D_ H H the Mangku Negoro , reigning Prince of Surakarta (Java) with his wife and child , who are now on their way to Holland . They will spend a few weeks in London _x000D_ 25 June 1926

Created by
TopFoto

Constructing ConConCor



how to 'annotate' a sample?

7 to 8 crowd workers per sample

*4: contentious, 3: not contentious,
1: I don't know*

"De vrouw tegenover hem was nog maar een meisje, twintig naar schatting.

Een nauwsluitend zwart manteltje en rok, witte satijnen blouse, een kleine, chique, zwarte toque, modieus gedragen op één oor.

*Ze had een mooi, **exotisch** gezichtje, mat-witte huid, groote bruine oogen, git-zwart haar.*

Ze rookte een sigaret in een langen houder.

Haar gemanicuurde handen hadden donkerroode nagels."

"The woman opposite him was only a girl, estimated to be twenty.

A close-fitting black cloak and skirt, white satin blouse, a small, chic, black toque, worn fashionably on one ear.

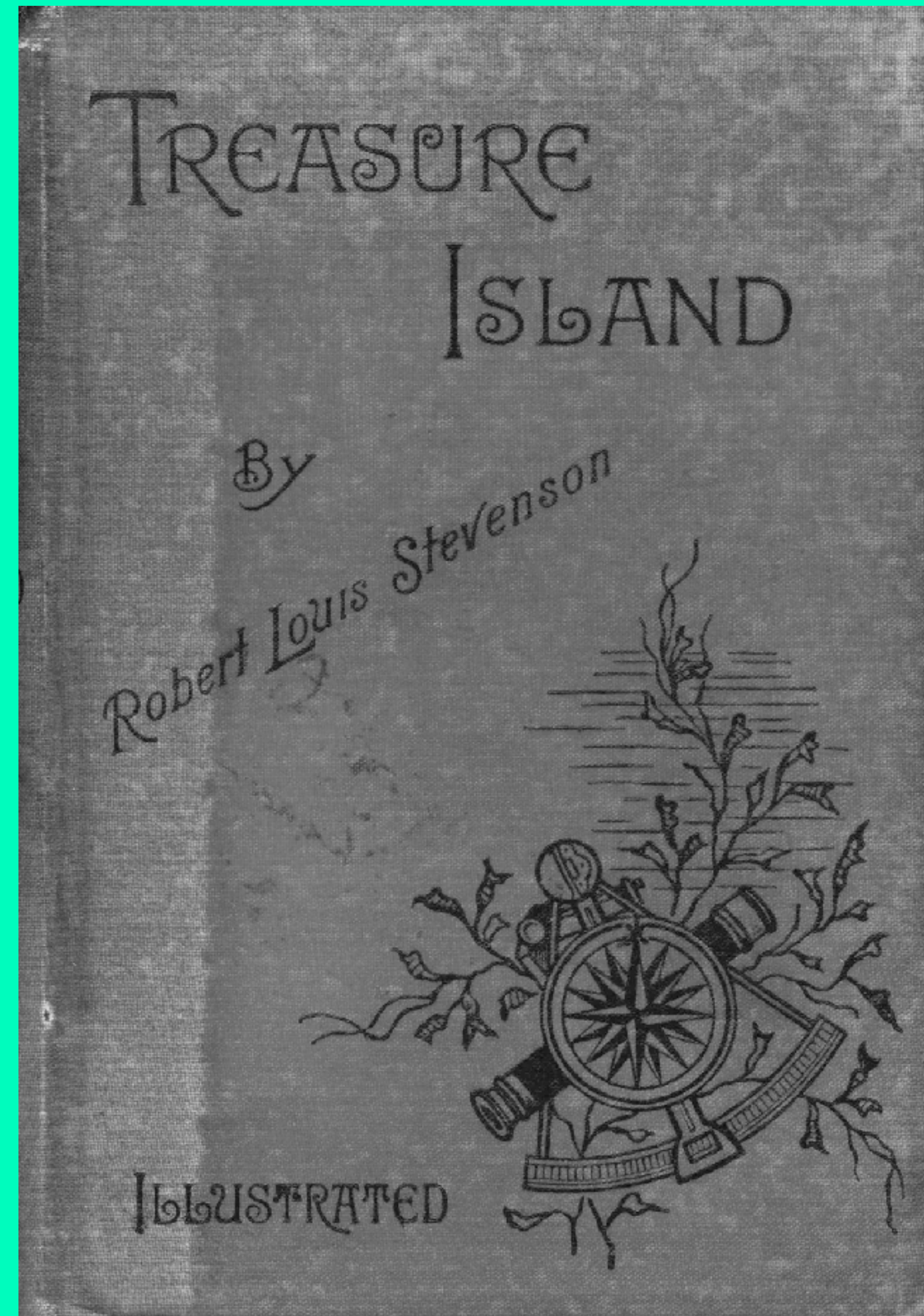
*She had a pretty, **exotic** face, matte white skin, big brown eyes, jet-black hair.*

She smoked a cigarette in a long holder.

Her manicured hands had dark red nails."

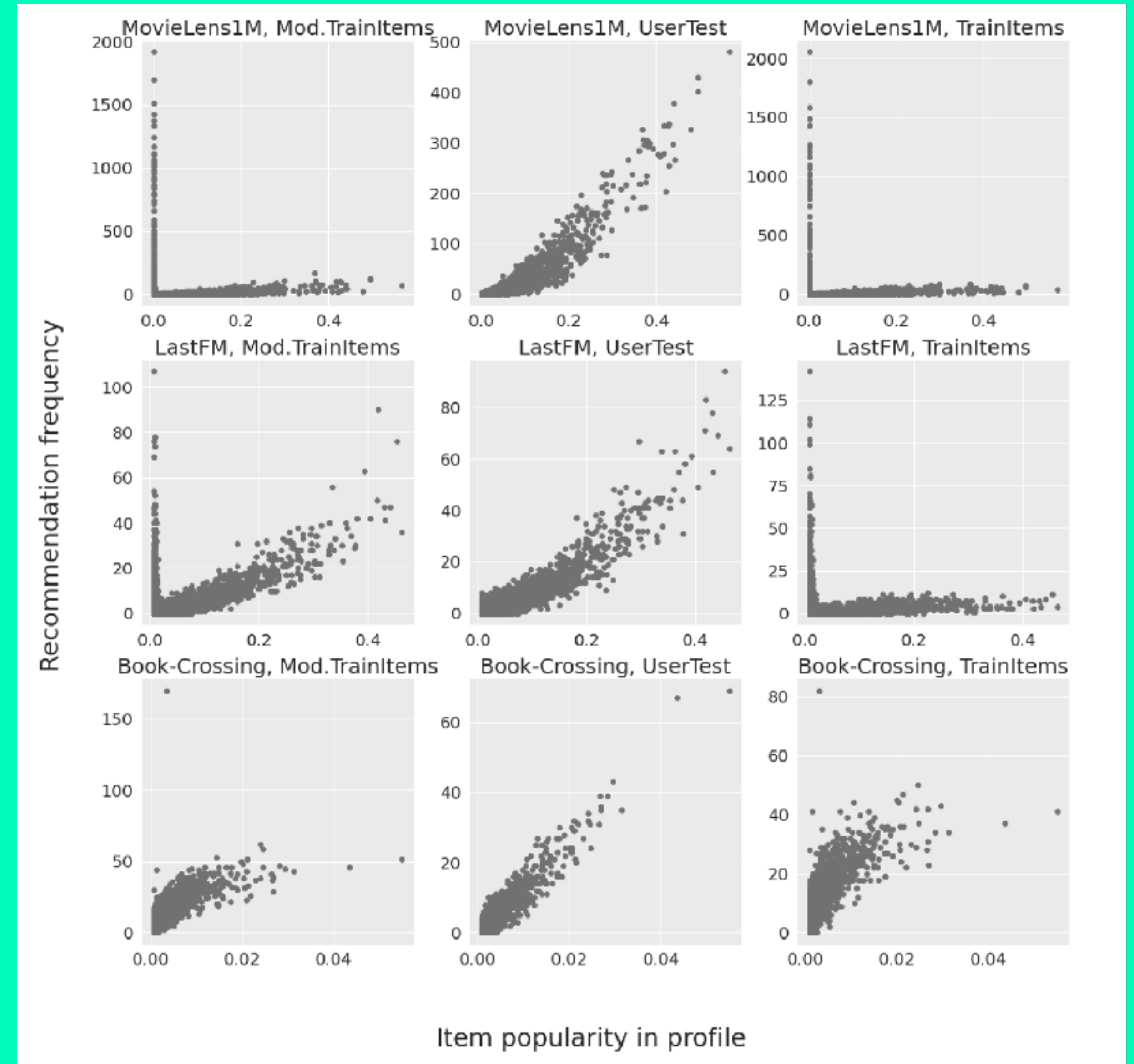
detecting problematic stereotypes
in books

*savage, primitive, lazy, conniving,
superstitious, treacherous, wily,
crafty, inscrutable, docile,
backwards, slant-eyed, naked*



your preferences are not your
neighbour's preferences

ethics in book recommendation
bias in recommendations in
general



what algorithm is best suited to my problem?

interpretability

transparency

resource use



Presentation and usage of polyvocal knowledge

SABIO

1M

API

https://sabio.diginfra.net/api/v1/

QUERY

RESULTS

1,675 / 146,694 (1.1%)

OBJECT METADATA

Keywords

plantage

Start date

1/01/01

End date

2017/01/01

Classification

Select...

Department

Select...

INDICATOR

Engine

TypicalityEngine/v0

Minimum score

0.0%

Maximum score

100%

VOCABULARY

Preset

Colonialism

Terms

bewindhebber,bewindvoerder,bomba,bombay,cimarron,derde wereld,dwerg,expeditie,gouverneur,halfbloed,hottentot,inboorling,indiaan,indisch,indo,inheer

OBJECT DETAILS



AUKANERS (BOSNEGERS)

* 74.5%

ca. 1865

46355

Zeven kleine aquarellen met verschillende bevolkingsgroepen in passepartout: Indianen met het wapen van Suriname (zie nr. -a); koopvrouwen te Paramaribo (b); plantagearbeidster (s); Caraïben (Indianen; g); misi met kind in zondagspak (q); kleurlingdametje te Paramaribo (r); Aukaners (Bosnegers; v). Oorspronkelijk behoorden deze zeven kleine aquarellen tot een grote kaart van de rivier de Suriname, in 1877 gemaakt door G.P.H. Zimmerman (hij heeft de metingen verricht en tekeningen gemaakt in 186...

Read more

VIEW SOURCE

INDICATOR SCORES

nense

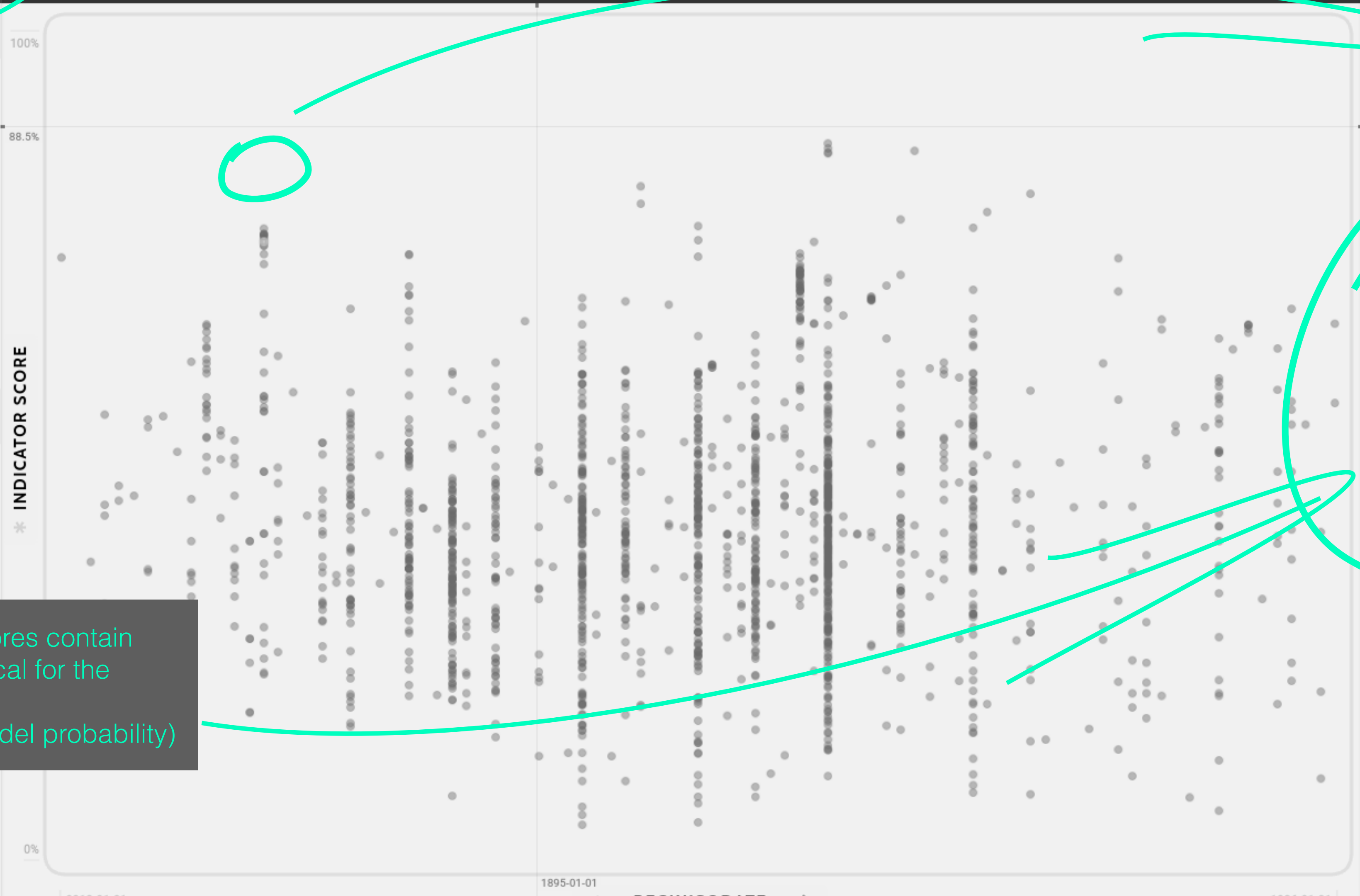
98.5

ge'lee

98.5

keyword search term (1675 matches)

objects with high scores contain language that is typical for the collection (~ high language model probability)



Alter Heritage

presenting alternative terms

involve different user groups

integrating with museum systems

Instructies
<

Herstellen #1

<

1

2

3

4

5

6

>

✓✓✓✓✓✓

Alle objecten zijn ingediend. Je kunt de taak nu afronden door dit venster te sluiten. Of je kunt wijzigingen aanbrengen in je bewerkingen en antwoorden en ze opnieuw indienen. Bedankt voor je deelname!

Je kunt je bewerkingen indienen nadat je de verplichte * vragen hieronder hebt beantwoord.

In hoeverre was naar jouw mening de representatie van LHBTQ+ personen in dit object inclusief **VOOR** je bewerkingen, waarbij 1 volledig niet-inclusief is en 10 volledig inclusief? *

1

2

3

4

5

6

7

8

9

10

In hoeverre is naar jouw mening de representatie van LHBTQ+ personen in dit object inclusief **NA** je bewerkingen, waarbij 1 volledig niet inclusief is en 10 volledig inclusief? *

1

2

3

4

5

6

7

8

9

10

Wat zijn je beweegredenen voor het bewerken van de metadata van dit object? *

Uitleg te geven aan een offensieve term.
 Een meer neutraal term te gebruiken in de metadata - vooral om meer objecten te kunnen vinden.
 Misschien is een metadata tag "LGBTQI+" or "queer" nuttig om erbij te zetten maar misschien is dat te generiek.

Wil je andere bewerkingen uitvoeren die niet mogelijk zijn in deze interface? Zo ja, wat zijn deze bewerkingen?

Als je nog iets wilt delen, laat dit dan hier achter.

Titel

Ontdekking van sodomie in een klooster te Brugge, 1578

■

"sodomie" is een bijbelse term dat refereert aan een homoseksuele ACT (sorry - ik ken de NLse woord er voor niet)

✕

⚠️

Mogelijke offensieve of kwetsende term

✕

Beschrijving

Arrestatie van de Franciscaner monniken in een klooster te Brugge op beschuldiging van sodomie, 18 mei 1578. Onder begeleiding van soldaten worden de monniken uit het klooster afgevoerd. Onder hen de beruchte Broer Cornelis van Dordrecht. Met onderschrift van 15 regels in het Duits. Linksonder genummerd: 145. De prent maakt deel uit van een album.

Onderwerp: Wat

1578-05-18 - 1578-05-18

Cornelis, Broer

+

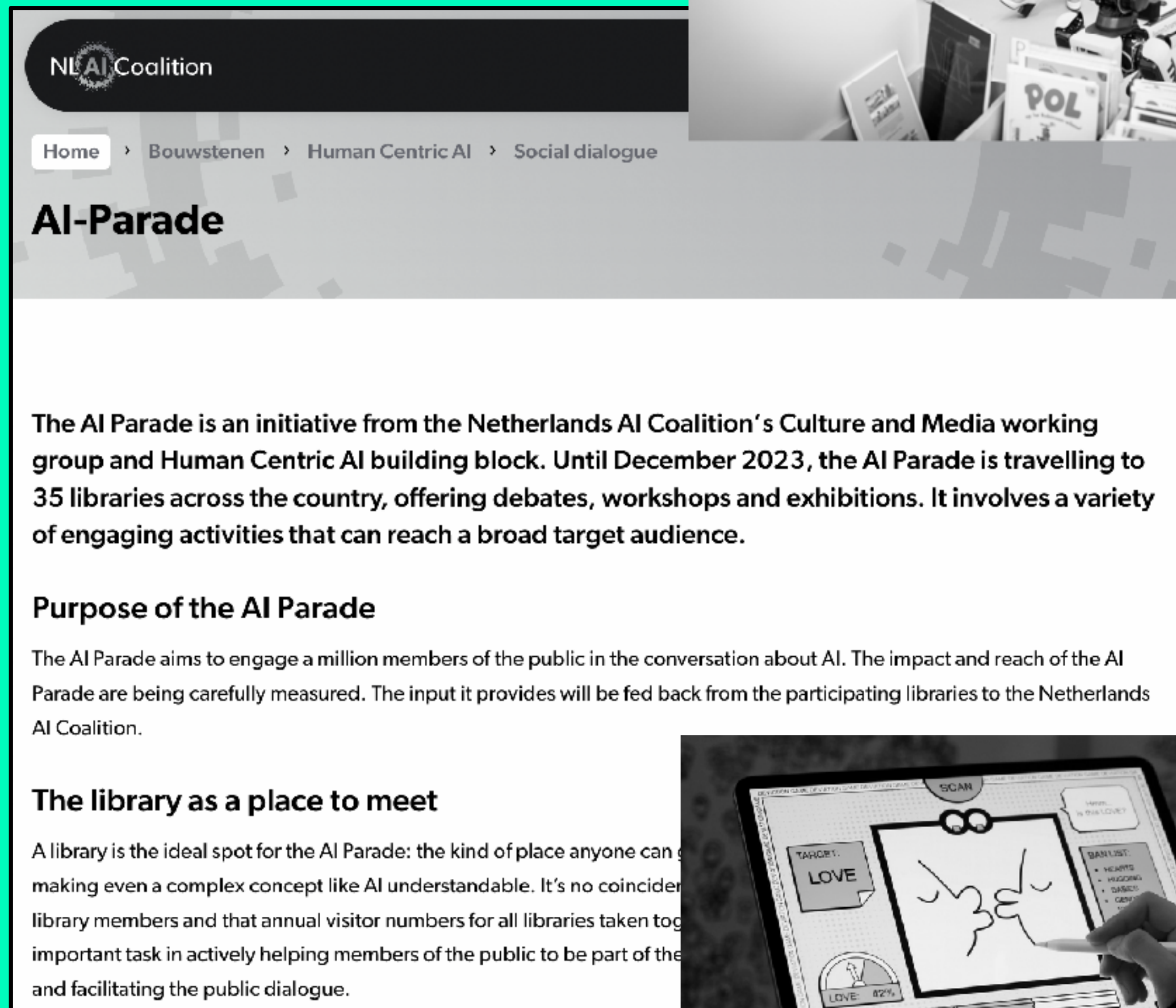
BETTER AI IS A
COMMUNITY EFFORT

AI education

for all ages

for all levels

for all domains



The screenshot shows the website of the NL AI Coalition. The header includes the NL AI Coalition logo and a navigation menu with links to Home, Bouwstenen, Human Centric AI, and Social dialogue. The main heading is "AI-Parade". Below this, a paragraph states: "The AI Parade is an initiative from the Netherlands AI Coalition's Culture and Media working group and Human Centric AI building block. Until December 2023, the AI Parade is travelling to 35 libraries across the country, offering debates, workshops and exhibitions. It involves a variety of engaging activities that can reach a broad target audience." This is followed by a section titled "Purpose of the AI Parade" which explains the initiative's goal to engage a million members of the public in the conversation about AI. The final section, "The library as a place to meet", describes libraries as ideal spots for the AI Parade due to their accessibility and high visitor numbers, emphasizing their role in facilitating public dialogue. To the right of the text is a black and white photograph of a group of people gathered around a small robot in a library setting, with a sign in the background that reads "AI-Parade".

NL AI Coalition

Home > Bouwstenen > Human Centric AI > Social dialogue

AI-Parade

The AI Parade is an initiative from the Netherlands AI Coalition's Culture and Media working group and Human Centric AI building block. Until December 2023, the AI Parade is travelling to 35 libraries across the country, offering debates, workshops and exhibitions. It involves a variety of engaging activities that can reach a broad target audience.

Purpose of the AI Parade

The AI Parade aims to engage a million members of the public in the conversation about AI. The impact and reach of the AI Parade are being carefully measured. The input it provides will be fed back from the participating libraries to the Netherlands AI Coalition.

The library as a place to meet

A library is the ideal spot for the AI Parade: the kind of place anyone can go to, making even a complex concept like AI understandable. It's no coincidence that libraries are meeting places for library members and that annual visitor numbers for all libraries taken together are high. Libraries play an important task in actively helping members of the public to be part of the conversation and facilitating the public dialogue.



data stewardship

consider where your data goes
ask how your data is used



broaden your horizon

*what works for you may not work
for your neighbour*



AI requires interdisciplinary teams



THIS PRESENTATION IS TEAM WORK



THANK YOU

and a big thank you to our funders: the Dutch Research Council, the European Union, and Dutch Digital Heritage Network

TRIFECTA has received funding from the European Union's FP7, Horizon 2020 and Horizon Europe research and innovation programmes under grant agreement 10188548. Cultural AI has received funding from the Dutch Research Council (NWO), the Dutch Digital Heritage Network (NDE) and Europeana.



Image sources:

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<https://www.axios.com/2025/04/22/ai-climate-toll-imf-study>

<https://www.vice.com/en/article/v7dg8m/ethical-ai-trained-on-reddit-posts-said-genocide-is-okay-if-it-makes-people-happy>

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<https://aibusiness.com/computer-vision/sony-scientists-find-computer-vision-models-biased-on-race-skin-color#close-modal>

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[ixlib=rb-4.1.0&rect=0%2C20%2C4500%2C2970&q=20&auto=format&w=320&fit=clip&dpr=2&usm=12&cs=strip](https://doi.org/10.1016/j.inffus.2023.101896)

Seven requirements for trustworthy AI. <https://doi.org/10.1016/j.inffus.2023.101896>

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